



Roll No:

**BPHARM**  
**(SEM. V) THEORY EXAMINATION 2025-26**  
**INDUSTRIAL PHARMACY I- THEORY**

**TIME: 3 HRS**

**M.MARKS: 75**

**Note:1.** Attempt all Sections. If require any missing data; then choose suitably.

**SECTION A**

**1. Attempt all questions in brief.**

**10 x 2 = 20**

|    |                                                                          |
|----|--------------------------------------------------------------------------|
| a. | Define pre-formulation studies and state its objectives.                 |
| b. | What is polymorphism? How does it affect drug stability?                 |
| c. | Write the significance of BCS classification in formulation development. |
| d. | Define granulation and mention its types used in tablet manufacturing.   |
| e. | What are the advantages of hard gelatin capsules?                        |
| f. | Define is otonicity. Why is it important in parenteral preparations?     |
| g. | What are the essential requirements of ophthalmic preparations?          |
| h. | Write the role of propellants in pharmaceutical aerosols.                |
| i. | What are the functions of surfactants in emulsions?                      |
| j. | Mention any four materials used in pharmaceutical packaging.             |

**SECTION B**

**2. Attempt any two parts of the following:**

**2 x 10 = 20**

|    |                                                                                                                                      |
|----|--------------------------------------------------------------------------------------------------------------------------------------|
| a. | Explain pre-formulation studies and discuss various physicochemical properties of drug Substances affecting formulation development. |
| b. | Describe different granulation methods used in tablet formulation. Add a note on Compression problems during tablet manufacturing.   |
| c. | Explain formulation and manufacturing considerations of syrups, suspensions and emulsions with suitable examples.                    |

**SECTION C**

**3. Attempt any five parts of the following:**

**7 x 5 = 35**

|    |                                                                                      |
|----|--------------------------------------------------------------------------------------|
| a. | Discuss BCS classification of drugs and its importance in drug formulation.          |
| b. | Explain tablet coating. Discuss types of coating and defects in coating.             |
| c. | Describe production and quality control tests of hard gelatin capsules.              |
| d. | Write a short note on formulation and preparation of pellets and their applications. |
| e. | Explain formulation of parenteral products and discuss aseptic processing.           |
| f. | Describe formulation and evaluation of ophthalmic preparations.                      |
| g. | Write the formulation and preparation method of cold cream and vanishing cream.      |